

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 2nd Year 1st Term Examination-2019
Basic Mechanical Engineering
ME 2119

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) Define thermodynamics. What are the scopes of thermodynamics? 07
- 1(b) What are the intensive and extensive properties of a system? Differentiate between homogeneous and heterogeneous system 10
- 1(c) What is meant by thermodynamic equilibrium? Explain mechanical, chemical and thermal equilibrium. 10
- 1(d) Explain energy transfer by work clarifying signs of work transfer. 08
- 2(a) Deduce the expression of PdV works for various quasi-static processes. 15
- 2(b) What is PMM1? Why it is impossible? 08
- 2(c) A piston-cylinder device with air at an initial temperature of 30°C undergoes an expansion process for which pressure and volume are related as given below: 12

P (kPa)	100	37.9	14.4
V (m ³)	0.1	0.2	0.4

Calculate the work done by the system.

- 3(a) Define the refrigerator and heat pump. Explain the concept of thermal energy reservoir. 10
- 3(b) State Kelvin Planck and Clausius statement for the second law. Explain the equivalence of these laws. 10
- 3(c) Air flows steadily at the rate of 0.5 kg/s through an air compressor, entering at 7m/s velocity, 100kPa pressure and 0.95 m³/kg volumes and leaving at 5m/s, 700 kPa and 0.19 m³/kg. The internal energy of the air leaving is 90 KJ/kg greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of 58 kW. 15
- (i) Compute the rate of shaft work input to the air in kW.
- (ii) Find the ratio of the inlet pipe diameter to the out pipe diameter.
- 4(a) What is air standard cycle? Define the following terms: 10
- (i) TDC (ii) BDC (iii) MEP (iv) Clearance volume and (v) displacement volume
- 4(b) Show that the efficiency of the otto cycle depends only on the compression ratio. 12
- 4(c) In an air standard diesel cycle, the compressor ratio is 15. Compressor begins at 0.1MPa, 40°C. the heat added is 1.675 MJ/Kg. Find: 13
- (i) The maximum temperature of the cycle (ii) the cycle efficiency and (iii) the temperature at the end of the expansion and (iv) the cut off ratio (v) the maximum pressure of the cycle.
- [For air, Cp= 1.005, Cv= 0.718, R=0.287 KJ/kg.k]

SECTION-B

5(a)	Define boiler. What are the essential features of a good boiler?	13
5(b)	Make a classification of steam boiler.	10
5(c)	Write the function of followings: (i) Blow off cock (ii) Fusible plug (iii) superheater and (iv) Economizer	12
6(a)	Draw and explain the working principle of Babcock and Wilcox boiler.	17
6(b)	A Lancashire boiler generates 2400 kg of dry steam per hour at a pressure of 11 bar. The grate area is 3 m ² and 90 kg of coal is burnt per m ² of grate area per hour. The calorific value of the coal is 33180 KJ/kg and the temperature of feed water is 17.5°C. Determine: (i) Actual evaporation per kg of coal. (ii) Equivalent evaporation from and at 100°C. (iii) Factor of evaporation and (iv) Efficiency of the boiler.	18
7(a)	Define engine. Differentiate IC and EC engine.	06
7(b)	Explain the working principle of a four stroke cycle diesel engine.	10
7(c)	Mention the difference between two stroke and four stroke engine. Which gives more power? Why?	10
7(d)	Briefly explain: (i) Flywheel (ii) crankshaft (iii) connecting rod.	09
8(a)	Define steam turbine. Explain working principle of an impulse turbine with neat sketch.	13
8(b)	Draw pressure velocity diagram of impulse turbine and reaction turbine.	10
8(c)	Explain the following terms: (i) SFC (ii) Brake power (iii) Detonation and (iv) Supercharging.	12

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 2nd Year 1st Term Examination-2019
Accounting and Industrial Law
Hum 2119

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section including question No. 4 from Section-A in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) What is accounting? 05
 1(b) How can you use your accounting knowledge? 10
 1(c) Who are the users of accounting information? 10
 1(d) What is the basic accounting equation? Identify and describe the steps in the accounting cycle/process. 10

- 2 On July 1, 2018, Mr. Enam established ENAM TRAVEL AGENCY. The following transactions were completed during the month:

2018 July 1	Invested Tk. 1,00,000 cash to start the agency.
2018 July 2	Paid Tk. 4000 cash for July office rent.
2018 July 3	Purchased office equipment for Tk. 25,000 cash.
2018 July 4	Incurred Tk. 3,000 of advertising costs in the "Daily Purbanchal" on account.
2018 July 5	Paid Tk. 6,000 cash for office supplies.
2018 July 6	Earned Tk. 95,000 for service rendered: Tk. 10,000 is received from customers and Tk. Balance of Tk. 85,000 is billed to customer on account.
2018 July 7	Withdraw Tk. 2,000 cash for personal use.
2018 July 8	paid "Daily Purbanchal" amount due in transaction 4.
2018 July 9	Paid employee salaries Tk. 22,000
2018 July 10	Received Tk. 85,000 in cash from customer who have previous been billed in transaction 6.

Instructions:

- (i) Journalise above the transactions in the book of MR. ENAM. 20
 (ii) Prepare a tabular analysis of the transactions using the following column headings: 15
 Cash; Accounts Receivable; Supplies; Office Equipment; Accounts Payable; and Enam's Capital.
- 3(a) What is ledger? What is the objective of preparing ledger? 10
 3(b) Define adjustment. Describe the objectives of adjusting entries. 05
 3(c) From the following Ledger balances of IFAZ DELIVERY SERVICES, prepare a trial balance on July 31, 2017, with the accounts arranged and fill in the missing amount for cash? 20

	Taka		Taka
Accounts receivable	8642	Repair Expense	961
Accounts payable	8396	Service Revenue	10610
Cash	?	Ifaz, Drawing	700
Delivery Equipment	49360	Ifaz, Capital	44636
Gas and Oil Expense	758	Salaries Expense	4428
Insurance Expense	523	Salaries payable.....	815
Notes payable	21450		
Prepaid Insurance	1968		

- 4(a) The adjusted trial balance for Pioneer Advertising Agency is presented. Prepare a statement of Comprehensive Income, Owner's Equity Statement and Financial Position as on October 31, 2018 (One month): 35

PIONEER ADVERTISING AGENCY
Adjusted Trial Balance, October 31, 2018

Account Titles	Debit (Taka)	Credit (Taka)
Cash	15,000	
Accounts Receivable	200	
Advertising Supplies	1,000	
Prepaid Insurance	550	
Office Equipment	5,000	
Accumulated Depreciation-Equipment		40
Notes payable		5,300
Accounts Payable		2000
Unearned Revenue		800
Salaries Payable		1,200
Interest Payable		50
Owner's Capital		10,000
Owner's Withdrawn	500	
Service Revenue		10,600
Salaries Expense	5,200	
Advertising Supplies Expense	1,500	
Rent Expense	900	
Insurance Expense	50	
Interest Expense	50	
Depreciation Expense	40	
	29,990	29,990

SECTION-B

- | | |
|---|----|
| 5(a) Define industrial law. | 05 |
| 5(b) State the importance of industrial law. | 10 |
| 5(c) Who is a certifying surgeon? | 05 |
| 5(d) Describe the responsibilities of the certifying surgeon. | 15 |
| 6(a) What are the precautions to be taken in case of fire in a factory? | 10 |
| 6(b) Discuss the rules regarding the employment of young persons on dangerous machines. | 10 |
| 6(c) How the compensation is distributed for injury or death caused by an accident? | 15 |
| 7(a) What is meant by conciliation? | 05 |
| 7(b) Distinguish between strike and lock-out. | 10 |
| 7(c) Who will pay the compensation in case of insolvency of the employer? | 10 |
| 7(d) What is meant by CBA? | 10 |
| 8(a) Define 'wages' as per the Bangladesh Labor Act, 2006. | 10 |
| 8(b) What are the deductions may be made from wages? | 15 |
| 8(c) Describe the payment procedure of unpaid wages of dead workers. | 10 |

Department of Leather Engineering
Khulna University of Engineering & Technology
 B.Sc. Engineering 2nd Year 1st Term Regular Examination, 2019
 Ch 2119
Physical Chemistry

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) What is molar conductance? Show graphically the variation of molar conductance against $\sqrt{c}$ for LiCl, NiSO_4 and Oxalic acid. Why does the molar conductance of weak electrolyte increase abnormally at infinite dilution? 12
- 1(b) With graph, explain the weak acid with strong base conductometric titration curve. Write down the cause of initial minima of the curve. What are the advantages of conductometric titrations over ordinary titrations? 12
- 1(c) State Kohlrausch's law of ionic mobilities. The molar conductance at 25°C of HCl, NaCl and CH_3COONa are 426.1, 126.4 and $91.0 \Omega^{-1}\text{cm}^2 \text{mol}^{-1}$ respectively at infinite dilution. Calculate the molar conductance of CH_3COOH at infinite dilution. 11
- 2(a) Show the interrelationship of E_{cell}° , ΔG° and K. From the relation explain when the reaction will be spontaneous, non-spontaneous or equilibrium. 08
- 2(b) Construct a cell of the followings with proper notation and cell reaction; and find out the emf of the cell 12
- (i) Cr(s) , $\text{Cr}^{3+}(\text{1M})$; $\text{H}^+(\text{1M})$, $\text{H}_2(\text{1atm})$, Pt(s)
 (ii) Ag(s) , $\text{Ag}^+(\text{1M})$; $\text{H}^+(\text{1M})$, $\text{H}_2(\text{1atm})$, Pt(s)
 (iii) Ag(s) , $\text{Ag}^+(\text{1M})$; Cr(s) , $\text{Cr}^{3+}(\text{1M})$
- Calculate the K value of Cr-Ag cell. Establish the graphical relation between E_{cell} and $\log K$ for Cr-Ag cell and indicate from the graph when the cell can do more work or less work.
- 2(c) What is glass electrode? How can it be used in measuring of pH of a solution? 08
- 2(d) Write a short note on 'Zeta potential'. 07
- 3(a) Illustrate a schematic diagram of lead acid battery. Write down the charge and discharge reaction of the battery. How does the battery develop voltage? 10
- 3(b) Draw an electrochemical cell containing three electrode systems. Write down the function of counter electrode, working electrode and reference electrode. 10
- 3(c) Explain reversible, quasi reversible and irreversible cyclic voltammogram with suitable diagram. 10
- 3(d) Write a short note on 'supporting electrolyte'. 05
- 4(a) Explain briefly fluorescence and chemiluminescence. 06
- 4(b) What is quantum yield? What are the causes of high and low quantum yield? 09
- 4(c) Deduce Langmuir adsorption isotherm and discuss the equation for the limiting condition of very low and high pressure. 12
- 4(d) "Adsorption is a exothermic process"- briefly illustrate this statement. 08

SECTION-B

- 5(a) Define solution with proper example. Why compounds is not a solution? 08
- 5(b) Derive a mathematical equation which correlates the lowering of vapour pressure with the molecular weight of solute. 11
- 5(c) Explain- why the vapour pressure of a solution is always less than of its solvent. 06
- 5(d) A current of dry air was bubble through a solution containing 7.50g solute in 750.0g water and then through pure water. The loss in mass of the solution was found to be 2.81g and that of water was 0.954g. Calculate the molecular weight of solute. 10
- 6(a) What do you mean by corrosion? Discuss the factors that enhance the rate of corrosion in Khulna region. 08
- 6(b) Briefly describe the different classes of electro-chemical cell in which corrosion takes place. 09
- 6(c) What is salting out effect? Discuss the effect of pH on the rate of underwater corrosion. 09
- 6(d) What is sacrificial anode? Describe the principle of prevention of corrosion by electrochemical method. 09
- 7(a) What is coagulation and peptization? Give suitable example and discuss the role of electrolyte on coagulation. 10

7(b)	What is CMC? Explain why-ionic micelles show high conductance.	10
7(c)	Write the essential differences between lyophilic and lyophobic colloids with example.	08
7(d)	Discuss the applications of colloids in our daily life.	07
8(a)	Briefly explain the solution process of solid-liquid solution.	10
8(b)	What do you mean by disperse phase and dispersion medium? Distinguish among true particle, colloidal particle and suspended particle.	10
8(c)	What is the molarity of a solution prepared by dissolving 75.5g of pure KOH in 540 ml of solution?	07
8(d)	Write a short note on 'pitting corrosion'.	08

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 2nd Year 1st Term Examination-2019
Mathematics III
Math 2119

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

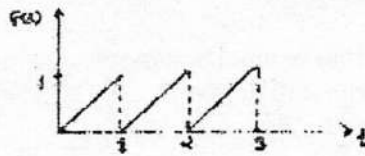
- 1(a) Define scalar point function and vector point function with example. 06
- 1(b) For the space curve, $x = t$, $y = t^2$, $z = 2/3t^3$ find (i) the curvature (ii) the torsion (iii) unit tangent vector and Normal vector. 17
- 1(c) The acceleration of a moving particle is given by $\frac{d^2\vec{r}}{dt^2} = 6t\hat{i} - 24t^2\hat{j} + 4\sin t\hat{k}$ such that initial ($t = 0$) velocity and displacement are given [velocity $\frac{d\vec{r}}{dt} = -\hat{i} - 3\hat{k}$ and displacement $\vec{r} = 2\hat{i} + \hat{j}$ at $t = 0$ sec]. Find acceleration, velocity and displacement at $t = 7$ sec. 12
- 2(a) Test the nature of the vector point function $\frac{M\vec{r}}{r^2}$ (by using vector operator ∇), where M is constant mass and $\vec{r}$ is position vector. 15
- 2(b) Define Gradient and Curl. Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$. 09
- 2(c) Evaluate ∇r^n . Hence find the value of ∇r^2 where $\vec{r}$ is the position vector of a point, what is the value of ∇r^2 at the point $(2, 1, 2)$. 11
- 3(a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ from $(0,0,0)$ to $(1,1,1)$ along the path C: The straight line joining $(0,0,0)$ to $(1,0,0)$; then $(1,1,0)$ and then $(1,1,1)$ where $\vec{F} = 2y\hat{i} - z\hat{j} + x\hat{k}$. 12
- 3(b) Evaluate $\int_S \vec{F} \cdot \vec{n} ds$, where S is the surface of the plane $2x + y + 2z = 6$ located in the first octant and $\vec{F} = (x + y^2)\hat{i} - 2xy\hat{j} + 2yz\hat{k}$. 12
- 3(c) Evaluate $\int_V \nabla \cdot \vec{F} dv$ where $\vec{F} = (2x^2 - 3z)\hat{i} - 2xy\hat{j} - 4x\hat{k}$ and v is the closed region bounded by the planes $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, $z = 1$. 11
- 4(a) What is the necessary condition of a function to be expressed as a Fourier Series? What do you mean by Dirichlet condition and Parseval's Identity? 06
- 4(b) Express the function $f(x)$ as a Fourier Series, where $f(x) = x$; $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$. Also analyze the Fourier Series of $f(x)$ graphically by taking partial sum of the series. 15
- 4(c) Discuss about expansion of half-range Fourier Series with figured. Hence find the Fourier Sine Series (with appropriate extension and show it in figure) of the following function. 14

$$g(x) = \begin{cases} \frac{2}{l}x & ; 0 < x < \frac{l}{2} \\ \frac{2}{l}(l-x) & ; \frac{l}{2} < x < l \end{cases}$$

SECTION-B

- 5(a) Define Laplace transform. If $f(t) = t^2$, $0 < t < 2$ and $F(t+2) = F(t)$, find $L\{F(t)\}$. 10
- 5(b) Find the following Laplace transform: 14
- (i) $L\{e^{3t}(2\cos 5t - 3\sin 5t)\}$
- (ii) $L\{F''(t)\}$ if $F(t) = te^{3t}$

- 5(c) Find $L\{F(t)\}$ where $F(t)$ is the periodic function as shown in the following figure, 11



- 6(a) State convolution property and using this, find the inverse Laplace transform of $\frac{1}{s^2(s+1)^2}$ 10

- 6(b) Find the following inverse Laplace transform, 12

(i) $L^{-1}\left\{\frac{2s^2 - 4}{(s+1)(s-2)(s-3)}\right\}$ by using Heaviside expansion formula.

(ii) $L^{-1}\left\{\frac{s+1}{s^2 + 6s + 25}\right\}$

- 6(c) Solve by using Laplace transform (any one): 13

(i) $4y'' - 4y' + 37y = 0; y(0) = 3, y'(0) = 10.5$

(ii) $Y'' + 9Y = \cos 2t$, if $Y(0) = 1, Y(\pi/2) = -1$

- 7(a) Reduce the following matrix to its canonical form and normal form. Hence find its rank. 17

$$[A : H] = \begin{bmatrix} 0 & 1 & 2 & \vdots & 3 \\ 2 & 3 & -1 & \vdots & 2 \\ 4 & 5 & 0 & \vdots & 1 \end{bmatrix} \text{ If } AX = H, \text{ Where } AH \text{ be the augmented matrix, then find the}$$

solution of X.

- 7(b) What is the necessary condition of a matrix to be invertible? If possible find A^{-1}, B^{-1} . Where 18

$$A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \text{ If } AX = H, \text{ Where } H = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \text{ then find the solution of}$$

X.

- 8(a) Discuss about the difference between matrixes and real number regarding arithmetic operator (+, -, ×, ÷) with example. 14

- 8(b) Define with example (i) Singular matrix (ii) Symmetric matrix (iii) Orthogonal matrix 09

- 8(c) Solve, $x_1 + x_2 + x_3 + x_4 = 0$ 12

$$x_1 + 3x_2 + 2x_3 + 4x_4 = 0$$

$$2x_1 + x_3 - x_4 = 0$$

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 2nd Year 1st Term Examination-2019
Leather Manufacturing Technology II
LE 2101

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

- | | |
|--|----|
| 1(a) State the principles of hair saving and hair burning unhairing. | 10 |
| 1(b) Write down the role of swelling in hair burning unhairing process. | 07 |
| 1(c) Write notes on the following unhairing technologies:
(i) Enzyme assisted chemical unhairing and (ii) Oxidative unhairing. | 10 |
| 1(d) What is painting? If you are given a raw goat skin how will you conduct soaking and painting? | 08 |
| 2(a) Define liming. Write the role of liming on leather manufacturing. | 08 |
| 2(b) Mention some common sharpening agents. Explain the chemistry of unhairing. | 08 |
| 2(c) The isoelectric point (IEP) of collagen shifts during liming-Explain. | 08 |
| 2(d) Write about the possible causes, effects and remedies of some common liming defects. | 11 |
| 3(a) State the mechanism of deliming using ammonium salts with their advantages and limitations. | 10 |
| 3(b) Write notes on the following deliming chemicals: (i) Acidic salts and (ii) Hydroxyl sinks. | 08 |
| 3(c) State the principles of carbon dioxide deliming with the pros and cons of the process. | 09 |
| 3(d) Write short notes on: (i) Lime blast (ii) Immunization of keratin | 08 |
| 4(a) Write down the objectives of bating. | 07 |
| 4(b) What are enzymes? Write about some common enzymes used in commercial bating preparations. | 08 |
| 4(c) Write about the factors influencing bating operation of leather manufacturing. | 14 |
| 4(d) Assume that you have a piece of delimed goat skin pelt and you have to produce very soft and stretchy final leather from it. Now formulate the suitable bating recipe for it. | 06 |

SECTION-B

- | | |
|--|----|
| 5(a) Define degreasing. Which types of chemicals are used in degreasing process? | 06 |
| 5(b) List out the solvents used in the degreasing process. What are the difficulties that arise with conventional solvent extraction degreasing process? How can it be overcome? | 12 |
| 5(c) Explain the mechanism of enzymatic degreasing process. | 10 |
| 5(d) If you are given a sheep skin when will you carry out degreasing? Also formulate a recipe. | 07 |
| 6(a) Pickling is nothing but the preparatory process for chrome tanning-Explain. | 07 |
| 6(b) Why acid swelling can be happened during pickling operation-Explain. How can it be overcome? | 08 |
| 6(c) Briefly explain the factors that should be considered in pickling process. | 10 |
| 6(d) If you are provided a bated pelt of goat origin, formulate a recipe for the production of wet blue leather. | 10 |
| 7(a) Write some features of tanning. | 04 |
| 7(b) Define basicity. Calculate the proctor's basicity of chrome liquor of 40% percentage basicity | 09 |
| 7(c) State the principles of single bath and double bath chrome tanning method. | 10 |
| 7(d) State the stereochemistry of Cr (III) complexes. | 07 |
| 7(e) Masking agent is needed in chrome tanning-Explain. | 05 |
| 8(a) Define basification. How chromium is fixed during basification process? | 07 |
| 8(b) Explain the reaction mechanism of basic chromium sulfate with collagen with necessary illustration. | 12 |
| 8(c) If a batch of wet blue leathers fail in boil test then what does it indicate? To pass this test what measures will you take? | 08 |
| 8(d) Write notes on the following terms: (i) Olation and (ii) Polymerization | 08 |